

Warm Up:

1. $\frac{1}{4} + \frac{3}{8}$

→ $\frac{2}{8} + \frac{3}{8} = \frac{5}{8}$

2. $4.5 - 5.2$

→ -0.7

3. $-14.5 - (-12.8)$

→ $-14.5 + 12.8 = -1.7$

4. $-2 + \frac{2}{3}$

→ $-\frac{6}{3} + \frac{2}{3} = -\frac{4}{3} = -1\frac{1}{3}$

5. On Friday morning, Hanna had \$235 in her checking account. On Friday afternoon, she used her debit card to spend \$42, \$10, and \$35. What is the balance in Hanna's account after these purchases?

→ $235 - (42 + 10 + 35) = 235 - 87 = \148

1.2 Homework Questions?

Speak Like A Mathematician:

additive identity: Zero – adding it to any number leaves the number unchanged.

multiplicative identity: One – multiplying any number by it leaves the number unchanged.

multiplicative inverses: Two numbers whose product is 1.

reciprocals: Another name for multiplicative inverses: a/b and b/a , where $a, b \neq 0$.

Key Concept — Properties of Equality

Reflexive Property	
Words:	Any quantity is equal to itself.
Symbols:	$a = a$
Examples:	$7 = 7$; $x + 3 = x + 3$

Symmetric Property	
Words:	If one quantity equals a second, the second equals the first.
Symbols:	If $a = b$, then $b = a$.
Examples:	If $9 = 4 + 5$, then $4 + 5 = 9$.

Transitive Property	
Words:	If one quantity equals a second and the second equals a third, the first equals the third.
Symbols:	If $a = b$ and $b = c$, then $a = c$.
Examples:	If $3 + 6 = 9$ and $9 = 4 + 5$, then $3 + 6 = 4 + 5$.

Example 1: Identify Properties of Equality

Heads up – the printed heading reads “Identity Properties of Equality,” but the task is to IDENTIFY which property is used.

- If $13 + 25 = 38$, then $38 = 13 + 25$.
→ Symmetric Property of Equality
- $y + 4 = y + 4$
→ Reflexive Property of Equality
- $22 + 7 = 22 + 7$
→ Reflexive Property of Equality
- If $36 = 17 + 19$, then $17 + 19 = 36$.
→ Symmetric Property of Equality

Example 2: Interpret Properties of Equality

- If the amount of sugar in a recipe is equal to the amount of flour plus 2 tablespoons, and the amount of flour plus 2 tablespoons is equal to the amount of milk, then the amount of sugar is equal to the amount of milk. Let a = the amount of sugar, b = the amount of flour plus 2 tablespoons, and c = the amount of milk.

A) Write an algebraic statement for this scenario.

→ If $a = b$ and $b = c$, then $a = c$.

B) Which property of equality makes this statement true?

→ Transitive Property of Equality

Example 3: Use Properties of Equality

- $y + 21 = \underline{\hspace{2cm}}$; Reflexive
→ $y + 21$ (a quantity equals itself)
- If $24 + 11 = 9 + 26$ and $9 + 26 = z$, then $24 + 11 = \underline{\hspace{2cm}}$;
Transitive
→ z

Key Concept — Addition and Multiplication Properties

Additive Identity	
Words:	For any number a , the sum of a and 0 is a .
Symbols:	$a + 0 = a$ and $0 + a = a$
Examples:	$13 + 0 = 13$

Additive Inverse	
Words:	A number plus its opposite is 0.
Symbols:	$a + (-a) = 0$
Examples:	$6 + (-6) = 0$

Multiplicative Identity**Words:** For any number a , the product of a and 1 is a .**Symbols:** $a \cdot 1 = a$ and $1 \cdot a = a$ **Examples:** $17 \cdot 1 = 17$ **Multiplicative Property of Zero****Words:** For any number a , the product of a and 0 is 0.**Symbols:** $a \cdot 0 = 0$ and $0 \cdot a = 0$ **Examples:** $42 \cdot 0 = 0$ **Multiplicative Inverse****Words:** For every nonzero number a/b , there is exactly one number b/a such that their product is 1.**Symbols:** $(a/b)(b/a) = 1$ **Examples:** $(3/4)(4/3) = 1$ **Associative Property****Words:** The way you GROUP three numbers when adding or multiplying does not change the result.**Symbols:** $(a + b) + c = a + (b + c)$; $(ab)c = a(bc)$ **Examples:** $(2 + 5) + 8 = 2 + (5 + 8)$ **Commutative Property****Words:** The ORDER in which you add or multiply does not change the result.**Symbols:** $a + b = b + a$; $ab = ba$ **Examples:** $9 + 4 = 4 + 9$

*Associative moves the parentheses; Commutative moves the numbers.
That one sentence separates the two properties students mix up most.*

Examples 4, 5, and 6: Evaluate Using Properties

- $4 - 2^2 + 8(2)$
→ $4 - 4 + 16 = 16$ (Additive Inverse: $4 - 4 = 0$)
- $(\frac{3}{2})(\frac{2}{3})(7)$
→ $1 \cdot 7 = 7$ (Multiplicative Inverse)
- $(4 - 3(2) + 7) \cdot 0$
→ 0 (Multiplicative Property of Zero – no need to simplify inside)

Example 7: Determine Which Property Was Used

- $114 + 71 + 19 + 26$
- $= 114 + (71 + 19) + 26$
→ Associative Property of Addition – the GROUPING changed
- $= 114 + 90 + 26$
→ Substitution (Simplify)
- $= 114 + 26 + 90$
→ Commutative Property of Addition – the ORDER changed
- $= 140 + 90$
→ Substitution (Simplify)
- $= 230$
→ Substitution (Simplify)

Examples 8 and 9: Evaluate Using Commutative and Associative

- $(\frac{5}{6})(9)(\frac{6}{5})$
→ $(\frac{5}{6})(\frac{6}{5}) = 1$, so the product is 9 (Commutative, then Multiplicative Inverse)
- Josefina needs to evaluate $(\frac{7}{3}) \cdot 2 \cdot 3$. How can she use the commutative property to make her evaluation easier?
→ Reorder to $(\frac{7}{3})(3)(2) = 7 \cdot 2 = 14$ – pair the 3 with the $\frac{7}{3}$ first.
- Jade buys groceries once each week and wants to know what she spent over 4 weeks: $32 + 27 + 28 + 33$. Use one of the properties to evaluate.
→ $(32 + 28) + (27 + 33) = 60 + 60 = \120
- Use the commutative and associative properties to evaluate $(\frac{5}{9}) \cdot 9 \cdot (\frac{9}{5}) \cdot 4$.
→ $(\frac{5}{9})(\frac{9}{5}) = 1$, leaving $9 \cdot 4 = 36$

Tonight's Homework: Paper Practice 1.3 (#1-12)

